A

SELECTED PROJECT PORTFOLIO
2026 · SLEMAN, DIY

DATA · SOFTWARE · COMPUTER VISION

Ahmad Maulana *Ibrahim*

Information Engineering undergraduate
Data and software projects

I turn raw data and camera footage into systems people can use: clear reports, reliable dashboards, and software that is easy to check.

MAIN PROJECTS · 04

- Highway Lane Occupancy & Hogging Detection
- Highway Lane Analytics — Data Pipeline & Dashboard
- NYC Mobility Lakehouse
- NusaCommerce Data Platform

SUPPORTING PROJECTS · 02

- Photobooth — Local-First Kiosk Prototype
- Rekberin — Transaction and Payment Contributions

SMALLER PROJECTS · 03

- Thrift Store / WantToSell
- River Water-Level Modeling
- OpenGL Porsche 911 Prototype

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ABOUT ME

I make data easier *to trust and use.*

3.41
GPA · SCALE NOT STATED
2024 — EXPECTED 2028

I am an Information Engineering student at Universitas Gadjah Mada. I enjoy building systems that turn messy data into something a team can understand and act on.

WHAT I DO

I build data flows, simple software, and tools people can check.

My work usually starts with a messy source: a database, a set of files, or a camera feed. I help turn it into something clearer: a clean table, a report, a dashboard, or a service another team can use.

I care about systems that are easy to understand and safe to run again. I also like showing the limits of a project honestly, so nobody expects more than what was actually built and tested.

EDUCATION

University	**Universitas Gadjah Mada**
Program	**Bachelor of Engineering, Information Engineering**
Study period	**2024 — expected 2028**
Location	**Sleman, DIY**

ORGANIZATIONAL EXPERIENCE

Find IT! UGM · Oct 2024 — May 2025	**Staff, Event Division**
Enforian UGM · Sep — Dec 2024	**Staff, Logistics & Catering**

AT A GLANCE

09 SELECTED PROJECTS	04 MAIN PROJECTS
30M+ TRIPS ANALYSED	109 AUTOMATED CHECKS

DATA TOOLS

Python · SQL · PySpark · Pandas · Parquet · Kafka · DuckDB · Delta Lake · Databricks

SOFTWARE & DATABASES

FastAPI · Next.js · Express · PostgreSQL · SQLite · MongoDB · SQLAlchemy · Prisma

VISION & MACHINE LEARNING

YOLO11n · ByteTrack · OpenCV · scikit-learn · CatBoost · XGBoost · PyTorch

INFRASTRUCTURE & TESTING

Docker Compose · GitHub Actions · pytest · Newman · Streamlit · Plotly

MAIN PROJECTS

01 Highway Lane Occupancy & Hogging Detection

02 Highway Lane Analytics — Data Pipeline & Dashboard

03 NYC Mobility Lakehouse

04 NusaCommerce Data Platform

SUPPORTING PROJECTS

01 Photobooth — Local-First Kiosk Prototype

02 Rekberin — Transaction and Payment Contributions

SMALLER PROJECTS

01 Thrift Store / WantToSell — Backend and Testing

02 River Water-Level Modeling — TMA Experiment Family

03 OpenGL Porsche 911 Prototype

MAIN PROJECTS / 01–02

From camera footage *to a dashboard.*

One project watches highway cameras and records what each vehicle does. The other collects those records and turns them into clear reports and charts.

Highway video → vehicle detection → lane tracking → event file → safe handoff → receiving service → queue → data store → dashboard

01 · CAMERA & LANES

Highway Lane Occupancy & Hogging Detection

A prototype that reads recorded highway video · one camera · six lanes

github.com/ibrahim119920/highway-lane-analytics

Portfolio page · michiportfolio.vercel.app

This project watches recorded highway video and works out which lane each vehicle is in, how long it stays there, and whether it stays in the passing lane longer than the rules allow. Each finding comes with a snapshot and a short video clip.

- The road is divided into six lanes on screen, so one setup can work with different camera resolutions.
- For large vehicles close to the camera, the system uses the bottom corner of the vehicle instead of the middle, so it does not jump between lanes.
- A lane change is only recorded after it is seen for several frames. A hogging signal is a rule match, not a legal violation.

158	111	8.65	109
VEHICLES FOLLOWED	EVENTS · 5-MIN VIDEO	FRAMES PER SECOND	AUTOMATED CHECKS

Python · YOLO11n · ByteTrack · OpenCV · pytest

02 · DATA PLATFORM

Highway Lane Analytics — Data Pipeline & Dashboard

A service that receives the events, checks them, stores them in layers, and shows the results

highway-lane-analytics-data-pipeline · private repository

Portfolio page · dashboard section

This project receives the event files, checks them, removes duplicates, and stores them in layers so they can be used for analysis. A dashboard then shows lane usage, vehicle types, hogging signals, and data quality.

- A secure receiving service accepts events in batches and rejects bad records instead of stopping the whole process.
- The same data can be sent again without double-counting. In testing, 111 events and 158 vehicle tracks were processed with no rejected records.
- The dashboard has four pages: overview, lane analysis, hogging explorer, and run quality.

111	158	0	4
EVENTS RECEIVED	VEHICLE TRACKS	RECORDS REJECTED	DASHBOARD PAGES

FastAPI · Kafka · DuckDB · Parquet · Streamlit · pytest

> **What to keep in mind.** This is a prototype using one camera and two short videos. Replaying old data is not live detection. A hogging signal is a rule match, not a violation. Vehicle counts are a lower bound because the tracker can lose an ID after a vehicle is blocked from view.

MAIN PROJECTS / 03–04

Large data, *handled carefully.*

Two projects about making large or recurring data safe to process, easy to check, and simple to reuse.

03 · LARGE DATA

NYC Mobility Lakehouse

Nine months of New York City taxi trips · about 30 million records

github.com/ibrahim119920/nyc-mobility-lakehouse

Portfolio page · michiportfolio.vercel.app

This project processes nine months of New York City taxi trip data — more than 30 million trips — and turns it into clean summary tables for analysis.

- The same monthly job can be run again without adding duplicate data.
- Data fingerprints and version history show that a rerun produces the same result.
- The summary tables match the detailed data with zero difference, then support 12 analysis questions about time, place, payment type, and trip length.

30.0M	9	12	0
TRIPS PROCESSED	MONTHS LOADED	ANALYSIS QUESTIONS	SUMMARY DIFFERENCE

Databricks · PySpark · SQL · Delta Lake · Medallion · replaceWhere

04 · PLATFORM DESIGN

NusaCommerce Data Platform

A design for moving online-store data from the order database to reports and dashboards

github.com/ibrahim119920/nusacommerce-data-platform

Portfolio page · michiportfolio.vercel.app

This project designs a data platform for an online store: data comes from the order database, passes through clear layers, and ends in reports and dashboards.

- The first working path loads new orders only, in small pages, and remembers where it stopped.
- Each load is saved only after it is complete, so a failed run can be retried safely.
- Data quality and freshness targets are part of the design, not an afterthought.

3	2	15	1
SOURCES DESIGNED	DATA AGREEMENTS	MINUTE LOOKBACK	ACTIVE LOADER

Python · PostgreSQL · SQLAlchemy · Data architecture · Data quality · Docker Compose

> **What was not built yet.** NYC covers January–September 2024 only, so it is not a full-year result. NusaCommerce currently runs the order-loading path only; the payment connection and other data sources are design targets. The analysis is descriptive, not causal.

SUPPORTING PROJECTS / 01–02

Software for *everyday operations.*

One photo booth that runs on a single device; one marketplace with a careful payment flow.

01 · LOCAL SOFTWARE

Photobooth — Local-First Kiosk Prototype

A self-service photo booth that keeps its data on one device and connects to a camera

Private repository

Portfolio page · michiportfolio.vercel.app

This is a photo booth prototype for one device. The screen, the local database, and the saved photos all live on the same machine, while a small service connects to the camera.

- Only one photo session can be active at a time, so two customers cannot accidentally share a session.
- Commands are safe to repeat, every important action is logged, and the local data can be checked or backed up.
- The camera connection can be tested without a physical camera, which makes the rest of the flow easier to develop.

Choose package → pay & verify → take photos → save locally → upload queue → result QR

1	Local	C++	Fake
ACTIVE SESSION	DATABASE	CAMERA CONNECTOR	CAMERA TESTING

Next.js · SQLite · C++ bridge · Transactions · Audit logs · SSE

02 · PAYMENTS

Rekberin — Transaction and Payment Contributions

A team marketplace for game accounts · my work focused on transactions and payment

github.com/aufaanggara/rekberin

Portfolio page · michiportfolio.vercel.app

A marketplace where people buy and sell game accounts. My contribution focused on the transaction and payment steps, from reserving a listing to confirming a payment.

- I built transaction services and made sure a listing cannot be reserved twice.
- Only the buyer and seller in a transaction can change its status.
- I connected QRIS payments in test mode and checked that payment notifications really came from the provider.
- Tests were added through three merged pull requests: #28, #38, and #44.

Listing → offer & chat → QRIS payment → payment check → handover → payout simulation

3	QRIS	2 h	1×
MERGED PULL REQUESTS	PAYMENT TEST MODE	BUYER CONFIRMATION	LISTING RESERVATION

TypeScript · Next.js · Prisma · PostgreSQL · Midtrans · REST APIs

What to keep in mind. The photo booth's physical camera was not verified in the saved project snapshot; testing used a simulated camera. Rekberin's payment integration uses a test account, and seller payouts and refunds are simulated.

SMALLER PROJECTS / 01–03

Smaller builds, *the same care.*

Shorter project work that still shows careful thinking about data, testing, and presentation.

01 · Online store backend

Thrift Store / WantToSell

A backend for an online thrift store, covering product listings, favourites, orders, reviews, and stock reservations.

- One item per product can be reserved safely, so two buyers cannot take the same item.
- Ten simultaneous purchase attempts produced one order and nine conflicts.
- Automatic tests were added; the saved report records one failed check.

JavaScript · Express · MongoDB · Mongoose · Newman

github.com/ibrahim119920/web-thrift-store

Portfolio page · michiportfolio.vercel.app

02 · Prediction experiments

River Water-Level Modeling

A group of experiments that predict river water levels, from simple seasonal comparisons to stronger machine-learning models.

- Models were tested in time order, so they could not learn from future data.
- Station and upstream-river information was included to improve the predictions.
- The work covers 30 stations and seven model families, with experiment notes for each run.

CatBoost · XGBoost · PyTorch · Pandas · Time series

github.com/ibrahim119920/TMA-CatBoostRegressor

Portfolio page · michiportfolio.vercel.app

03 · 3D graphics

OpenGL Porsche 911

A 3D rendering prototype of a Porsche 911, built in C++ with OpenGL for a graphics course project.

- Displays a 3D car model with lighting and surface detail.
- Loads model files and handles the window, camera, and basic interaction.
- Includes lighting and material effects for the car's appearance.

C++ · OpenGL · GLFW · GLAD · GLM · Assimp

Private repository

CLEAR WORK, HONEST LIMITS

Useful systems. *Straight answers.*

I am interested in data, software, and computer vision work. The projects above are prototypes and coursework; the notes say what was tested and what was not, so the work can be judged fairly.

Data flows · Online services · Computer vision · Data quality

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